

$$M\left(\int_{E_\delta} |g(x)|^q dx\right)^{1/q} < \varepsilon \Rightarrow \overline{\lim}_{k \rightarrow \infty} \left| \int_{E_\delta} f_k(x)g(x) dx \right| \leq \varepsilon.$$

所以 $\lim_{k \rightarrow \infty} \int_E f_k(x)g(x)dx = 0$.

(2) 若 $m(E) = \infty$, 令 $B_R = \{x \in \mathbb{R}^n \mid \|x\| \leq R\}$, $E_R = B_R \cap E$, 则 $m(E_R) < \infty$, 并有

$$\int_E |g(x)|^q dx = \lim_{R \rightarrow \infty} \int_{E_R} |g(x)|^q dx.$$

对于 $\forall \varepsilon > 0, \exists R_0 > 0$, 当 $R > R_0$ 时, 有

$$\begin{aligned} M\left(\int_{E \setminus E_R} |g(x)|^q dx\right)^{1/q} &< \varepsilon, \\ \left| \int_E f_k(x)g(x)dx \right| &\leq \left| \int_{E_R} f_k(x)g(x)dx \right| + \int_{E \setminus E_R} |f_k(x)g(x)|dx \\ &\leq \left| \int_{E_R} f_k(x)g(x)dx \right| + \varepsilon. \end{aligned}$$

令 $k \rightarrow \infty$, 即得结论.

13. $\forall A \in \mathfrak{M}_n$, 证明 $\int_A f(x)dx = 0$.

16. 记 $M = \sup_k \|f_k\|_\infty$, 于是对于 $\forall k$, $E(|f_k| > M)$ 是零测集. 因为 $f_k \xrightarrow{L^1} f$, 存在几乎处处收敛子列, 故 $|f(x)| \leq M$, a.e., 所以 $\|f\|_\infty \leq M$. 又 $\forall p > 1$, 有

$$\int_E |f(x)|^p dx = \int_E |f(x)||f(x)|^{p-1} dx \leq M^{p-1} \int_E |f(x)| dx = M^{p-1} \|f\|_1.$$

$$\Rightarrow f \in L^p(E) \cap L^\infty(E).$$

又

$$\begin{aligned} \int_E |f_k(x) - f(x)|^p dx &\leq \int_E |f_k(x) - f(x)|(|f_k(x)| + |f(x)|)^{p-1} dx \\ &\leq (2M)^{p-1} \|f_k - f\|_1 \rightarrow 0. \end{aligned}$$

18. 用 Schwarz 不等式.